UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Large Load Interconnection Reform
Docket No. RM26-4-000

Reply Comments of the Thermal Battery Alliance

I. Introduction

The Thermal Battery Alliance ("TBA") respectfully submits these Reply Comments in response to the Commission's Advance Notice of Proposed Rulemaking ("ANOPR") in Docket No. RM26-4-000. TBA represents companies developing and deploying thermal energy storage ("TES") systems capable of shifting industrial-scale electrical load away from periods of peak demand and toward hours of surplus generation. This flexibility enables large industrial facilities to interconnect to the transmission system without imposing the same peak-driven infrastructure needs as conventional firm load. In the case of hyper-flexible loads like TES, facilitating their quick interconnection and transmission service will support a more affordable grid, reindustrialization and American energy dominance.

The initial comments in this proceeding demonstrate substantial support for several points directly relevant to the Commission's inquiry. First, commenters overwhelmingly agree that large loads capable of providing durable, enforceable flexibility should be eligible for expedited interconnection pathways. Second, commenters further agree—explicitly or implicitly—that such expedited treatment is only feasible when the interconnecting load takes service on a non-firm or conditional basis. And third, commenters stress that the absence of a clear, Commission-directed, non-firm service option risks perpetuating unreasonable and inefficient outcomes, including the misallocation of upgrade costs to loads that do not contribute to system peaks.

These Reply Comments build on TBA's initial filing and several themes echoed throughout the record. TBA urges the Commission to require every jurisdictional transmission provider to offer at least one non-firm network transmission service option tailored to large flexible loads. TBA further provides a narrative framework describing how such a service should be structured—drawing in part on practical experience developed in the Southwest Power Pool ("SPP") stakeholder process and initial design concepts of the Price Adaptive Load Service ("PALS") proposal.

II. The Record Supports Requiring a Non-Firm Network Transmission Service for Large Flexible Loads

The initial comments in this proceeding reveal a consistent and compelling theme: when the Commission asserts jurisdiction over transmission-connected large loads, a significant number of stakeholders recommend that the Commission adopt reforms enabling expedited interconnection for loads capable of providing durable, asset-backed flexibility. This broad agreement spans regional transmission organizations ("RTOs"), transmission owners ("TOs"), public interest groups, storage developers, and flexible-load coalitions.

Commenters across the record emphasize that the existing framework, which treats all large loads as firm for both interconnection and transmission service, fails to recognize the operational reality of industrial facilities that are able and willing to curtail during system peaks. For example, the California Independent System Operator ("CAISO") explains that only loads that are "completely curtailable or dispatchable" should qualify for expedited interconnection study treatment, underscoring that the depth of flexibility determines whether a project can be evaluated without peak-driven upgrades. CAISO Comments at 13. CAISO further emphasizes that large flexible loads must provide real-time telemetry and component-level visibility akin to large generators. Id. at 11.

Stakeholders recognized the need for modelling designed to evaluate load flexibility. The New York Independent System Operator ("NYISO") explains that curtailment commitments from large loads provide a practical mechanism to avoid adverse resource adequacy impacts and reduce the need for system reinforcements. NYISO Comments at 8. NYISO also identifies the development of "Flexible Load models for Large Loads" as a priority in its ongoing internal work plan—a clear recognition that formal modeling constructs are needed to appropriately study and integrate flexible load.

SPP's comments provide perhaps the clearest illustration of workable non-firm constructs, as the RTO is already pursuing tariff revisions to create the Conditional High Impact Large Load Service ("CHILLS") and the Price Adaptive Load Service ("PALS"). SPP Comments at 6. These services allow large loads to interconnect quickly by accepting curtailment during peak or emergency periods and operating primarily when prices are low or when transmission capacity is unconstrained. CHILLS and PALS directly reflect the kind of non-firm service the Commission is evaluating in this proceeding, and they demonstrate the administrative feasibility of integrating large flexible loads through conditional service rights.

The Midcontinent Independent System Operator ("MISO") likewise highlights the importance of region-specific processes for integrating large loads, emphasizing that its existing tools "effectively facilitate interconnection of large loads" and urging the Commission to provide flexibility for regional approaches under the independent entity variation standard. MISO Comments at 5-6. While MISO's framework has worked to date, a specific non-firm service that allows MISO to curtail loads to avoid peaks would facilitate longer term growth. Conditional service is an appropriate mechanism for integrating flexible loads without incurring unwarranted upgrades.

PJM Interconnection, L.L.C. ("PJM"), urged caution and raised the need for an analysis of the trade-offs of the ANOPR's "connect-and-manage" framework. PJM Comments at 10-11. PJM's concerns center on ensuring that conditional service does not shift excessive operational risk into real-time operations. The Commission can address the concerns from PJM and others through clear guidance on minimum design features for non-firm service, so that RTOs can implement such models with confidence in their reliability safeguards. TBA recommends well-defined curtailment rules, telemetry requirements, and penalties for non-performance.

The views of state commissions reinforce this consensus. While state commenters emphasize the importance of respecting retail-jurisdictional boundaries and not allowing federal reforms to intrude upon local distribution service, they do not dispute that transmission-connected large loads fall within the Commission's jurisdiction. Their concerns relate to cost allocation and the avoidance of cost shifts to retail ratepayers—concerns that a properly designed non-firm network service directly addresses by ensuring that loads subject to curtailment are not treated as contributors to peak-driven infrastructure.

Taken together, the initial comments make clear that: (1) flexible large loads can and should be studied differently than inflexible loads; (2) non-firm network transmission service is the linchpin that makes expedited interconnection both feasible and reliable; and (3) a Commission requirement is needed to ensure that all transmission providers offer such a product, rather than relying on voluntary regional experimentation. The Commission should therefore require each jurisdictional transmission provider to establish a non-firm network transmission service option consistent with the guidelines discussed below.

III. A Guidelines-Based Requirement Is Appropriate and Necessary

The record demonstrates that a guidelines-based requirement is the most administratively efficient and legally sustainable approach for ensuring that large flexible loads have a workable transmission service pathway. *See* Comments of Electricity Customer Alliance at 14-16 (setting forth how a guidelines-based approach could work in practice). While many commenters express enthusiasm for integrating flexible load, they also acknowledge that the Commission must provide clear direction if non-firm service is to be implemented consistently across regions.

Adopting guidelines, as opposed to a *pro forma* tariff, has pros and cons, but is a good first step to support a variety of approaches in the near term. Guidelines provide more flexibility, which risks failing to provide adequately clear direction. At the same time, however, a guidelines-based approach facilitates positive regional variation. And most importantly, guidelines could be adopted expeditiously. In TBA's view, carefully tailored guidelines that set forth clear minimum standards for non-firm transmission service offerings would offer significant benefits. We recommend that the Commission take such action now, leaving room to adopt more precise guidance in the future if necessary as implementation takes shape.

Numerous RTOs and transmission providers emphasize the need for regional flexibility. MISO, for example, urges the Commission to preserve the independent entity variation standard and permit regional tailoring in recognition of differing system conditions and planning practices. MISO Comments at 5-6. PJM, while expressing reservations about certain expedited study concepts, similarly stresses that any conditional service model must be structured with clear reliability safeguards. PJM Comments at 10-11. These comments collectively support a federal framework that establishes minimum features without prescribing a uniform tariff.

Other commenters reinforce the need for federal direction to avoid inconsistency. Storage developers, flexible-load coalitions, and industrial customers explain that voluntary regional adoption has produced a patchwork of treatment, with only some RTOs—most prominently SPP—actively developing conditional service options. Without a Commission directive, they argue, large flexible loads will continue to face inefficient and uneven access to transmission service, undermining the goals of this proceeding.

Public interest organizations and academic economists further caution that the absence of clear federal standards can create incentives for inefficient system behavior, including load defection, over-investment in behind-the-meter resources, and regulatory arbitrage. They note that a properly structured non-firm service option would enable flexible loads to complement system needs by consuming energy during low-price periods and avoiding peaks, reducing both operating costs and long-term infrastructure expansion.

A guidelines-based approach could also mitigate concerns articulated in comments from state commissions. While these comments emphasize the boundaries of retail jurisdiction, even states who assert that jurisdiction over *interconnection* of large load customers is exclusively state-jurisdictional do not contend that *transmission service* frameworks are state-jurisdictional. Non-firm transmission service offerings are a necessary component of enabling expedited interconnection of large load customers, even if jurisdiction over the interconnection process remains with states. Further, many states—such as theVirginia State Corporation Commission—indicate that federal rules governing transmission-connected large loads can be compatible with state authority so long as they do not compel retail outcomes. Their concerns focus on cost allocation and the protection of retail customers, which a well-designed non-firm service directly addresses by ensuring that non-firm loads are not allocated costs based on peak contributions they do not make.

A guidelines-based requirement therefore achieves three objectives simultaneously: it ensures that large flexible loads have access to a meaningful non-firm transmission option in all regions; it preserves regional variation by allowing transmission providers to tailor implementation details; and it addresses reliability and cost-allocation concerns identified by RTOs, TOs, and state commissions. We recommend the Commission adopt such a framework in the final rule.

IV. Minimum Features of a Non-Firm Network Transmission Service

The record demonstrates broad support for the Commission establishing minimum features for a non-firm network transmission service tailored to large flexible loads. These features must ensure reliability, avoid unjust and unreasonable cost shifts, and allow transmission providers to model flexible loads accurately in interconnection and planning studies. Drawing on the comments submitted in this proceeding, as well as practical experience in SPP's stakeholder process on the Price Adaptive Load Service ("PALS"), TBA identifies three core elements of any non-firm network service: (A) pricing and cost allocation that reflect reduced upgrade needs; (B) eligibility criteria grounded in enforceable, asset-backed flexibility; and (C) modeling, scheduling, and operational treatment consistent with non-firm service status.

A. Pricing and Cost Allocation Must Reflect the Reduced Upgrade Requirements of Non-Firm Loads

As TBA explained in its initial comments, customers with non-firm withdrawal rights do not impose the same peak-driven infrastructure needs as firm network integration transmission service ("NITS") customers. This principle is broadly supported in the record. Numerous commenters—including storage developers, flexible-load coalitions, and several public interest groups—emphasize that large loads capable of curtailing during peak or emergency conditions should not be allocated upgrade costs as if they were fully firm.

The initial PALS proposal from SPP provides a concrete model. Under PALS, SPP staff have indicated that transmission charges will likely be based on volumetric use. This framework is workable so long as the volumetric charges are reflective of the relatively low cost causation from curtailable loads and proportionate to off-peak non-firm rates already in use. This approach aligns cost responsibility with actual system use.

Other commenters endorse similar principles. CAISO's comments, for example, implicitly recognize the need for differentiated treatment of loads based on their degree of flexibility by limiting expedited interconnection pathways to those that are "completely curtailable or dispatchable." CAISO Comments at 13. NYISO likewise suggests that curtailment commitments can mitigate resource adequacy risks that drive planning-level infrastructure needs. NYISO Comments at 8. These comments together support the conclusion that the Commission should require that non-firm loads not be assigned costs for upgrades based on hours in which they are contractually barred from taking service.

A well-structured non-firm network service also addresses state concerns regarding cost allocation. Several state commissions—including the Pennsylvania Public Utility Commission, the California Public Utilities Commission, and the Virginia State Corporation Commission—warn against cost shifts to retail customers. Because non-firm loads do not contribute to system peaks, creating the non-firm network service class allows for avoiding infrastructure and eliminates cost-shifts, directly addressing these concerns. Non-firm transmission customers can be exempted from peak-driven cost allocation without triggering cost-shift challenges.

B. Eligibility Must Require Asset-Backed Flexibility and Enforceable Curtailment Obligations

The second essential feature of any non-firm network service is an eligibility framework grounded in verifiable, asset-backed flexibility. Throughout the record, commenters emphasize that non-firm rights should not be offered to loads lacking the ability to curtail on a sustained basis. Instead, non-firm eligibility must be tied to tangible operational characteristics.

TBA's initial comments define this category as battery systems, inherently flexible industrial processes, or large loads backed by thermal energy storage. Commenters representing data centers, storage developers, and hybrid facilities support similar definitions, often referencing the need for enforceable curtailment capabilities. These positions are consistent with CAISO's requirement that only "completely curtailable or dispatchable" loads qualify for expedited treatment. CAISO Comments at 13.

The PALS design again offers a workable template. Under the initial PALS conceptual proposals, SPP or the Transmission Owner may curtail load "during critical peaking grid events," and the customer must comply immediately or face unreserved-use penalties. PALS loads are also subject to ramp-rate and ride-through requirements, demonstrating that large flexible loads can meet the same operational expectations as other controllable resources. These enforceable obligations address concerns raised by PJM and its transmission owners that conditional service may otherwise shift excessive operational risk into real-time operations.

Eligibility standards should therefore include: (1) documentation of asset-backed flexibility (e.g., thermal storage capacity, process flexibility, or supporting storage); (2) enforceable curtailment commitments reflected in the service agreement; (3) telemetry, data reporting, and performance requirements commensurate with the offered flexibility; and (4) penalties or reclassification for non-performance. These features provide the certainty RTOs and TOs require to plan and operate the system reliably.

C. Modeling, Scheduling, and Operational Treatment Must Align With Non-Firm Status

The third essential feature is that non-firm loads must be modeled in interconnection, transmission service, and planning studies consistent with their curtailment rights. Nearly all RTO commenters emphasize this need. NYISO identifies the development of "Flexible Load models for Large Loads" as a priority initiative for 2026, indicating that new study constructs are necessary. NYISO Comments at 3. CAISO emphasizes the importance of real-time telemetry and visibility into load behavior. CAISO Comments at 11. PJM stresses the need for greater clarity in eligibility for conditional service. PJM Comments at 10.

Under the SPP PALS model (as currently in conceptual development), transmission planners assume that the load will not operate during meaningful peak or emergency conditions. This assumption eliminates the need for Designated Network Resources and avoids infrastructure expansions triggered by coincident peak load. PALS loads operate primarily during low-LMP hours, subject to a bid cap likely tied to the marginal unit's fuel and O&M costs. This structure promotes efficient economic dispatch and mitigates uplift by increasing generator utilization and reducing the cycling of inflexible resources.

A Commission-directed non-firm network service should therefore require: (1) study assumptions that reflect curtailment commitments; (2) economic scheduling that allows loads to consume during low-cost hours within a defined operating envelope; (3) explicit tariff provisions governing curtailment priority relative to firm network, firm PTP, and non-firm PTP service; (4) consistent application of non-firm assumptions across interconnection, transmission service, and long-term planning; and (5) Commission guidance that non-firm loads do not create Planning Reserve Margin or resource adequacy obligations.

When these elements are combined, the result is a transmission service that preserves reliability, reduces unnecessary infrastructure spending, enhances system efficiency, and ensures just and reasonable cost allocation.

V. Reliability, Cost Allocation, and Jurisdiction

The comments submitted in this proceeding demonstrate that a properly structured non-firm network transmission service not only advances the Commission's goals of timely and efficient interconnection, but also strengthens system reliability and ensures fair cost allocation. PJM, CAISO, and NYISO each emphasize in their own ways that flexible loads must be integrated through mechanisms that provide clear operational visibility, enforceable curtailment rules, and predictable real-time behavior. CAISO's insistence on component-level telemetry for large flexible loads, combined with its requirement that only fully curtailable or dispatchable loads qualify for expedited study, underscores that reliability remains paramount. CAISO Comments at 11, 13. NYISO likewise notes that curtailment commitments from large loads can mitigate resource adequacy concerns and ensure that such loads do not adversely affect peak conditions. NYISO Comments at 8.

Cost allocation concerns raised by transmission owners and state commissions also support the adoption of non-firm service. The Indicated PJM Transmission Owners caution that the Commission's reforms must not shift costs to native load customers or undermine existing efforts to integrate large load customers. PJM TOs Comments at 2. State commissions—including the California Public Utilities Commission, the Virginia State Corporation Commission, and the Pennsylvania Public Utility Commission—likewise emphasize the need to avoid imposing peak-driven upgrade costs on customers that do not contribute to those peaks. These concerns are directly addressed when non-firm loads are exempted from peak-related cost allocation and priced in accordance with non-firm off peak PTP methodologies.

Finally, the record reflects broad agreement that a clear delineation between wholesale transmission service and state-jurisdictional retail service is indispensable to preserving the statutory balance embodied in the Federal Power Act. Several ISOs declined to respond to whether the proposed rules fall under FERC jurisdiction, but expressed that clarity resulting from the rulemaking will be beneficial. See, e.g., NYISO, Initial Comments at 3; CAISO, Initial Comments at 11–13. CPUC and VSCC argued that FERC does not have jurisdiction over large load interconnection and urged for the rulemaking to take a cooperative approach to solving the issues raised in the ANOPR. See, e.g., CPUC, Initial Comments at 1–2; VSCC, Initial Comments at 1–3. However, we continue to believe that providing non-firm network transmission service to transmission-connected large loads falls squarely within the Commission's jurisdiction over transmission in interstate commerce.

The record further demonstrates that clarifying the ability of thermal energy storage resources to purchase electricity at wholesale for conversion and resale as thermal energy does not erode state jurisdiction. As TBA and several public-interest commenters note, TES is a new form of storage technology that is capable of charging electrically and discharging either electrically or thermally, and therefore should be afforded the same wholesale transaction rights long recognized for battery storage and other energy-shifting resources. See TBA, Initial Comments at 8; ClearPath, Initial Comments at 6-9; L. Lynne Kiesling, Ph.D, Initial Comments at 11. Regulatory clarity on this point promotes parity across storage technologies, avoids inefficient investment and deployment decisions, and facilitates the efficient integration of thermal energy storage, while fully preserving state jurisdiction over retail matters. This approach ensures that large flexible loads can be reliably and efficiently integrated into the bulk power system without intruding upon state authority.

VI. Conclusion

For the reasons stated above, the Thermal Battery Alliance respectfully urges the Commission to require each jurisdictional transmission provider to adopt at least one non-firm network transmission service option for eligible large flexible loads. Such a requirement should include minimum features addressing pricing, eligibility, and modeling, while preserving regional flexibility for implementation. A guidelines-based approach will ensure consistency across regions, support system reliability, avoid unjust and unreasonable cost shifts, and unlock the full potential of large flexible loads—including those backed by thermal energy storage—to contribute to a more efficient and resilient grid.

Respectfully submitted,

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